

The Yard (Rail / Industrial / Logistics, Infrastructure, Armored Warfare)

Executive Summary

This is a modular, scalable, and flexible framework for building a realistic, immersive, and engaging simulation of a rail yard, industrial complex, and armored warfare environment. The framework is designed to be modular, allowing for the addition or removal of features as needed. The framework is designed to be scalable, allowing for the simulation of a small yard or a large industrial complex. The framework is designed to be flexible, allowing for the simulation of a wide variety of scenarios and environments.

Design Goals & Gameplay

Logistics & Infrastructure Warfare

The first priority is to ensure that the simulation is realistic and engaging. This means that the simulation must be able to handle a large number of objects and entities, and it must be able to simulate a wide variety of scenarios and environments.

Armored Industrial Combat

The second priority is to ensure that the simulation is realistic and engaging. This means that the simulation must be able to handle a large number of objects and entities, and it must be able to simulate a wide variety of scenarios and environments.

Logistics & Resources

Autonomy & Route/Decision Logic

Autonomous entities must be able to navigate a complex environment, avoiding obstacles and finding the most efficient route to their destination. This requires a sophisticated pathfinding algorithm and a system for managing the autonomy of the entities.

Sensor & Information Warfare

Entities must be able to detect and track other entities in the environment. This requires a sophisticated sensor system and a system for managing the information warfare between the entities.

Logistics & Resources

Entities must be able to manage their resources, including fuel, ammunition, and other supplies. This requires a sophisticated resource management system and a system for managing the logistics of the simulation.

Deployment & Progress

The simulation must be able to save and load game states, allowing for the deployment of the simulation and the progress of the simulation. This requires a sophisticated save/load system and a system for managing the progress of the simulation.

Spatial Constraints (Nreal AR)

Device Capabilities

Target: High-end mobile devices (e.g., AR/VR headsets, smartphones, tablets).

Playable Area & Scaling

100m x 100m physical area, 1 km² virtual area (1:10 scale). Designed for tabletop AR.

Tracking & Occlusion

Uses sensor fusion (IMU, camera) for movement tracking. Depth maps (if available) provide limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

100m x 100m grid (1 unit = 1 m). Origin (0,0) = southeast corner.

Key Locations

Base: 50m x 50m Area (Southeast Corner) - 0,0 to 50,50. Includes: Warehouse, Armored Vehicle, Armory, etc.

Rail Yard: 100m x 100m Area (Northwest Corner) - 100,0 to 200,100. Includes: Industrial Complex, Command Center, etc.

Central Area: 100m x 100m Area (Center) - 50,50 to 150,150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,0 to 150,50. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,50 to 150,100. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,100 to 150,150. Includes: Rail Line, etc.

Central Area - 100,150 to 150,200. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,150 to 150,200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,200 to 150,250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,250 to 150,300. Includes: Rail Line, etc.

Central Area - 100,300 to 150,350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,350 to 150,400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,400 to 150,450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,450 to 150,500. Includes: Rail Line, etc.

Central Area - 100,500 to 150,550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,550 to 150,600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,600 to 150,650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,650 to 150,700. Includes: Rail Line, etc.

Central Area - 100,700 to 150,750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,750 to 150,800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,800 to 150,850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,850 to 150,900. Includes: Rail Line, etc.

Central Area - 100,900 to 150,950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,950 to 150,1000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,1000 to 150,1050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,1050 to 150,1100. Includes: Rail Line, etc.

Central Area - 100,1100 to 150,1150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,1150 to 150,1200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,1200 to 150,1250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,1250 to 150,1300. Includes: Rail Line, etc.

Central Area - 100,1300 to 150,1350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,1350 to 150,1400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,1400 to 150,1450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,1450 to 150,1500. Includes: Rail Line, etc.

Central Area - 100,1500 to 150,1550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,1550 to 150,1600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,1600 to 150,1650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,1650 to 150,1700. Includes: Rail Line, etc.

Central Area - 100,1700 to 150,1750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,1750 to 150,1800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,1800 to 150,1850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,1850 to 150,1900. Includes: Rail Line, etc.

Central Area - 100,1900 to 150,1950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,1950 to 150,2000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,2000 to 150,2050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,2050 to 150,2100. Includes: Rail Line, etc.

Central Area - 100,2100 to 150,2150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,2150 to 150,2200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,2200 to 150,2250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,2250 to 150,2300. Includes: Rail Line, etc.

Central Area - 100,2300 to 150,2350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,2350 to 150,2400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,2400 to 150,2450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,2450 to 150,2500. Includes: Rail Line, etc.

Central Area - 100,2500 to 150,2550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,2550 to 150,2600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,2600 to 150,2650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,2650 to 150,2700. Includes: Rail Line, etc.

Central Area - 100,2700 to 150,2750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,2750 to 150,2800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,2800 to 150,2850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,2850 to 150,2900. Includes: Rail Line, etc.

Central Area - 100,2900 to 150,2950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,2950 to 150,3000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,3000 to 150,3050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,3050 to 150,3100. Includes: Rail Line, etc.

Central Area - 100,3100 to 150,3150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,3150 to 150,3200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,3200 to 150,3250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,3250 to 150,3300. Includes: Rail Line, etc.

Central Area - 100,3300 to 150,3350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,3350 to 150,3400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,3400 to 150,3450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,3450 to 150,3500. Includes: Rail Line, etc.

Central Area - 100,3500 to 150,3550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,3550 to 150,3600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,3600 to 150,3650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,3650 to 150,3700. Includes: Rail Line, etc.

Central Area - 100,3700 to 150,3750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,3750 to 150,3800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,3800 to 150,3850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,3850 to 150,3900. Includes: Rail Line, etc.

Central Area - 100,3900 to 150,3950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,3950 to 150,4000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,4000 to 150,4050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,4050 to 150,4100. Includes: Rail Line, etc.

Central Area - 100,4100 to 150,4150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,4150 to 150,4200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,4200 to 150,4250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,4250 to 150,4300. Includes: Rail Line, etc.

Central Area - 100,4300 to 150,4350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,4350 to 150,4400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,4400 to 150,4450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,4450 to 150,4500. Includes: Rail Line, etc.

Central Area - 100,4500 to 150,4550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,4550 to 150,4600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,4600 to 150,4650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,4650 to 150,4700. Includes: Rail Line, etc.

Central Area - 100,4700 to 150,4750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,4750 to 150,4800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,4800 to 150,4850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,4850 to 150,4900. Includes: Rail Line, etc.

Central Area - 100,4900 to 150,4950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,4950 to 150,5000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,5000 to 150,5050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,5050 to 150,5100. Includes: Rail Line, etc.

Central Area - 100,5100 to 150,5150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,5150 to 150,5200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,5200 to 150,5250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,5250 to 150,5300. Includes: Rail Line, etc.

Central Area - 100,5300 to 150,5350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,5350 to 150,5400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,5400 to 150,5450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,5450 to 150,5500. Includes: Rail Line, etc.

Central Area - 100,5500 to 150,5550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,5550 to 150,5600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,5600 to 150,5650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,5650 to 150,5700. Includes: Rail Line, etc.

Central Area - 100,5700 to 150,5750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,5750 to 150,5800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,5800 to 150,5850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,5850 to 150,5900. Includes: Rail Line, etc.

Central Area - 100,5900 to 150,5950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,5950 to 150,6000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,6000 to 150,6050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,6050 to 150,6100. Includes: Rail Line, etc.

Central Area - 100,6100 to 150,6150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,6150 to 150,6200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,6200 to 150,6250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,6250 to 150,6300. Includes: Rail Line, etc.

Central Area - 100,6300 to 150,6350. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,6350 to 150,6400. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,6400 to 150,6450. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,6450 to 150,6500. Includes: Rail Line, etc.

Central Area - 100,6500 to 150,6550. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,6550 to 150,6600. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,6600 to 150,6650. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,6650 to 150,6700. Includes: Rail Line, etc.

Central Area - 100,6700 to 150,6750. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,6750 to 150,6800. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,6800 to 150,6850. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,6850 to 150,6900. Includes: Rail Line, etc.

Central Area - 100,6900 to 150,6950. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,6950 to 150,7000. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,7000 to 150,7050. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,7050 to 150,7100. Includes: Rail Line, etc.

Central Area - 100,7100 to 150,7150. Includes: Warehouse, Armory, etc.

Coal & Lumber Piles - 100,7150 to 150,7200. Includes: Coal Piles, Lumber Piles, etc.

Warehouse Area - 100,7200 to 150,7250. Includes: Warehouse, Armory, etc.

Rail Line Corridor - 100,7250 to 150,7300. Includes: Rail Line, etc.

Central Area - 100,7300 to 150,7350. Includes: Warehouse, Armory, etc.